
UniVer: A Unified Perspective for Multi-step and Multi-draft Speculative Decoding

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Abstract

Speculative decoding accelerates Large Language Models via draft-then-verify, where verification can be framed as an Optimal Transport (OT) problem. Existing approaches typically handle multi-draft and multi-step aspects in isolation, applying either flat OT to single-step drafts or per-token rejection sampling to tree-structured candidates. This separation leaves the joint regime (where multi-step dependencies meet multi-draft branching) poorly optimized, as local verification rules fail to exploit the coupling between horizontal and vertical dimensions of candidate trees. In this paper, we propose a unified perspective that casts tree-based verification as a conditional OT problem. Our key insight is that vertical dependencies can be abstracted through prefix acceptance probabilities, which act as dynamic scaling factors to actively guide horizontal draft selection. Based on this principle, we introduce UniVer, a verification algorithm that jointly optimizes across tree levels by composing local optimal transport plans under prefix constraints. We prove that UniVer remains lossless and achieves the optimal acceptance rate under the proposed conditional framework. Extensive experiments across different tasks and models demonstrate that UniVer improves acceptance length by 4.2% to 8.5% over standard recursive rejection sampling without replacement, while maintaining exact distributional alignment with the target model.

1 Introduction

Modern Large Language Models (LLMs) suffer from high inference latency due to their autoregressive nature. Speculative decoding [13, 3] mitigates this problem without compromising output quality: a lightweight draft model predicts future tokens, which are then verified by the target model in parallel. From an algorithmic perspective, a fundamental challenge in speculative decoding is to design an optimal verification strategy that maximizes the expected acceptance rate while maintaining statistical fidelity to the original model (*i.e.*, ensuring the output distribution is identical to that of vanilla autoregressive decoding).

Recent studies improve acceptance rates by extending the naive (local) verification strategy along two orthogonal dimensions: optimizing horizontal selection among multiple candidates and managing vertical dependencies across multiple steps. On one hand, multi-draft methods such as SpecTr [21] and Greedy method [8] select multiple draft tokens at each step and optimize the acceptance rates from an Optimal Transport (OT) perspective. However, these approaches remain vertically "myopic", treating each generation step as an independent event and failing to account for the sequential dependencies in tree-structured drafts. On the other hand, multi-step methods [20, 24] focus on optimizing verification vertically across sequences or trees. Yet, these methods are horizontally limited: they rely on local,

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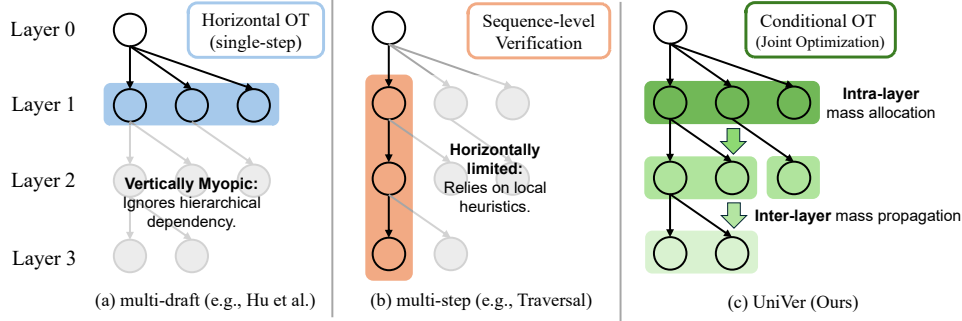


Figure 1: Conceptual illustration of verification paradigms. **(a) Multi-Draft approaches** (e.g., Hu et al. [8]) optimize horizontal selection via OT but ignore vertical dependencies (*vertically myopic*). **(b) Multi-Step approaches** (e.g., Traversal Verification [24]) optimize vertical acceptance but rely on local heuristics for horizontal selection (*horizontally limited*). **(c) UniVer** unifies both dimensions via Conditional OT, enabling parallel intra-layer mass allocation and inter-layer mass propagation.

per-token rejection sampling, which is suboptimal compared to coordinated, multi-draft selection. Consequently, existing paradigms suffer from a structural trade-off: they either optimize across multiple candidates at a single depth or across multiple steps for a single candidate, but fail to achieve joint optimality across both dimensions (Figure 1).

This discrepancy raises a fundamental question: *How to unify multi-draft selection and multi-step coordination into a single, principled verification framework?* From our perspective, the core difficulty lies in the combinatorial complexity of a candidate tree, where the exponential growth of potential paths makes direct joint optimal transport highly non-trivial. To bypass this complexity, we shift the focus from solving the global OT problem to a recursive local-global coordination. By introducing the *prefix acceptance probability* of a node v (denoted as $\tilde{p}_v$) as the dynamic scaling factor, we actively reformulate horizontal verification into a conditional optimal transport problem. This enables a unified strategy that couples multi-draft selection with cross-level dependencies while maintaining strict losslessness.

Guided by this principle, we propose UniVer (Unified Verification), a principled framework that decomposes tree-structured verification into a top-down allocation phase and a post-order decision phase. Unlike prior approaches [15, 24] that rely on Recursive Rejection Sampling without replacement (RRSw), UniVer enables parallel computation of acceptance plans for all branches at the same depth. By leveraging conditional OT, our algorithm determines the acceptance mass of child nodes within a single probability space, effectively eliminating the sequential dependency inherent in RRSw. As a result, UniVer provides a lossless yet efficient paradigm for tree speculative decoding. Empirical evaluations demonstrate that UniVer consistently outperforms single-step OT methods such as Greedy method, and achieves up to a 8.5% increase in acceptance length over Vanilla RRSw.

Our contributions are summarized as follows:

- **Unified Framework.** We propose a framework named UniVer, which unifies multi-step and multi-draft speculative decoding. By identifying prefix acceptance probabilities p_v as dynamic scaling factors, we reformulate tree-based verification as a series of conditional optimal transport problems.
- **Theoretical Foundation.** We prove that UniVer achieves the conditional optimum for any $p_v \in (0, 1]$ under a specific sampling strategy (*i.e.*, Greedy selection [8]), attaining the theoretical upper bound of acceptance under prefix constraints. We further demonstrate that this bound is strictly non-inferior to that of the Greedy method, establishing UniVer as a principled generalization of single-step OT optimality to multi-step tree verification.
- **Empirical Validation.** UniVer streamlines verification into a single-pass computation by eliminating the sequential dependencies inherent in RRSw and Traversal. Extensive experiments across various LLMs and tree architectures show that UniVer consistently improves acceptance length by 4.2% to 8.5% over standard RRSw under temperature 1.0 while maintaining losslessness.

2 Preliminaries

2.1 Speculative Decoding and OT

Let $\{\mathcal{M}_b, \mathcal{M}_s\}$ be a pair of probabilistic models over vocabulary Σ . For any given accepted prefix $\mathbf{x}$, we refer to $\mathcal{M}_b(\cdot | \mathbf{x})$ as the target distribution and $\mathcal{M}_s(\cdot | \mathbf{x})$ as the draft distribution, respectively.

Single-Draft Case. In standard speculative decoding [13, 3], a single draft token $x \sim \mathcal{M}_s(\cdot | \mathbf{x})$ is verified against the target distribution $\mathcal{M}_b$. The acceptance rate is determined by the probability of alignment between $\mathcal{M}_s$ and $\mathcal{M}_b$. Formally, this corresponds to an optimal transport problem, where the optimal acceptance rate α^* is given by the total variation distance (*i.e.*, the maximum achievable overlap between the target and draft distribution):

$$\alpha^* = \sum_{x \in \Sigma} \min\{\mathcal{M}_b(x | \mathbf{x}), \mathcal{M}_s(x | \mathbf{x})\}. \quad (1)$$

Local Heuristics for Multi-Draft. When n draft tokens are generated, a straightforward extension is to apply the single-token verification sequentially. Methods such as Recursive Rejection Sampling (RRS) and its without replacement variant (RRSw) [17, 4, 9, 27] adopt this strategy: they verify candidates one by one using local pairwise comparisons between $\mathcal{M}_b$ and $\mathcal{M}_s$, re-normalizing the residual distribution after each rejection. While RRSw prevents redundant sampling of identical tokens, it remains a *local heuristic* that processes candidates independently rather than coordinating their acceptance globally.

Multi-Draft with Optimal Transport. In contrast, some recent works [21, 8, 11] frame multi-draft verification as an OT problem. Hu et al. [8] propose a Greedy method. They first deterministically select $m - 1$ drafts with highest probabilities from $\mathcal{M}_s$, then sample the final draft from the residual $\mathcal{M}_s^-$. Under this sampling strategy, they derive a closed-form verification strategy that achieves the theoretical upper bound of the acceptance rate:

$$\alpha_{\text{Greedy}}^*(\mathcal{M}_b, \mathcal{M}_s) = \sum_{i \in \text{Top}_{m-1}(\mathcal{M}_s)} \mathcal{M}_b(i) + \sum_{i \in \Sigma} \min\{\mathcal{M}_b(i), \mathcal{M}_s^-(i)\}, \quad (2)$$

where $\mathcal{M}_s^-$ denotes the residual draft distribution. Unlike RRSw, this OT-based approach works under the above sampling structure (Top _{$m-1$} plus one residual sample) and coordinates the acceptance of all m candidates in a unified probability space. Under such sampling scheme, Eq. (2) maximizes the marginal acceptance probability at each step.

2.2 Multi-step Strategies

Consider a draft model generating a candidate tree T . For clarity, we use $\mathbf{v} = (v_0, \dots, v_\gamma)$ to represent a root-to-node chain $\mathbf{v} \subseteq T$, where v_0 denotes the root node and the chain length is $\gamma + 1$.

Local Heuristics. The baseline approach applies vanilla verification at each step independently. For any fixed chain $\mathbf{v} = (v_0, \dots, v_\gamma)$, let the acceptance ratio be $r(v_i) = \mathcal{M}_b(v_i) / \mathcal{M}_s(v_i)$ at position v_i . Under local heuristics, the acceptance probability is computed independently at each position as $\min\{r(v_i), 1\}$. Consequently, the actual acceptance probability $p_{v_k}^{\text{Local}}$ of each prefix chain of $\mathbf{v}$ is given by the product of individual, locally truncated probabilities:

$$p_{v_k}^{\text{Local}} = \prod_{i=1}^k \min\{r(v_i), 1\}, \quad 1 \leq k \leq \gamma.$$

This *post-hoc aggregation* treats each verification step as independent, failing to exploit the coupling of probability mass along the sequence, which leads to suboptimal cumulative acceptance rates.

Block Verification. For single-chain decoding, Sun et al. [20] propose Block Verification. Unlike local verification, which independently truncates the local density ratio at 1.0 before accumulation, Block Verification computes the prefix acceptance probability $p_{v_k}^{\text{Block}}$ by first performing cumulative multiplication and then applying truncation:

$$p_{v_k}^{\text{Block}} = \min\left\{r(v_k) \cdot p_{v_{k-1}}^{\text{Block}}, 1\right\}, \quad 1 \leq k \leq \gamma,$$

where $p_{v_0}^{\text{Block}} = 1$. This mechanism enables a high density ratio ($r(v_k) > 1$) to recharge the prefix probability diminished by prior steps. In contrast, local verification follows $p_{v_k}^{\text{Local}} = \min\{r(v_k), 1\} \cdot p_{v_{k-1}}^{\text{Local}}$, which permanently penalizes the chain for any local mismatch. Consequently, Block Verification yields a strictly non-inferior cumulative acceptance probability, as $p_{v_k}^{\text{Block}} \geq p_{v_k}^{\text{Local}}$ holds for all $1 \leq k \leq \gamma$.

Traversal Verification. Traversal Verification [24] extend Block Verification to tree structures through a *post-order traversal* strategy. This approach leverages the joint probability for vertical optimization, but handles the horizontal dimension via RRSw. Specifically, Traversal Verification processes the tree bottom-up by iteratively verifying candidates against the residual target distribution, falling back to sibling nodes or the parent upon rejection. In other words, it lacks joint optimization across siblings, failing to exploit the horizontal coupling.

3 Method

3.1 Overview

To bridge the gap between local horizontal selection and suboptimal vertical coordination in existing methods, we present UniVer, a unified framework for multi-draft and multi-step speculative decoding. As presented in Figure 2, UniVer operates through a two-stage verification process that jointly optimizes horizontal and vertical optimization.

Allocation Phase: Top-Down Probability Propagation. Starting from the root node with initial probability $\tilde{p}_{\text{root}} = 1$, we propagate the acceptance mass layer by layer. For each layer, the acceptance plans for all independent branches are computed in parallel. For each node v , we treat $\tilde{p}_v$ as the *effective prefix acceptance probability* and compute the conditional optimal transport plan between the scaled target distribution $\tilde{p}_v \cdot \mathcal{M}_b(\cdot | v)$ and the draft distribution $\mathcal{M}_s(\cdot | v)$. This yields marginal acceptance probabilities $p_v(u)$ for each child $u \in \mathcal{C}(v)$, which are then normalized into conditional acceptance probabilities $\tilde{p}_u$ for the next layer. This propagation continues until all leaf nodes are processed.

Decision Phase: Post-Order Tree Traversal. With all acceptance probabilities pre-computed, we perform a standard post-order tree traversal (as defined in Traversal Verification [24]). For each node $v \in T$, we draw $\eta \sim U(0, 1)$ and accept v if $\eta < \tilde{p}_v$ (leaf) or $\eta < \tilde{p}_v^{\text{res}}$ (non-leaf). If a node is rejected, we proceed to the next node in traversal order until an acceptance or all nodes are rejected. Upon acceptance at a leaf, we sample the next token from $\mathcal{M}_b$; upon acceptance at a non-leaf node, we sample from the residual distribution.

Algorithm 1 UniVer

Require: Draft tree T with root r , target model $\mathcal{M}_b$, draft model $\mathcal{M}_s$
Ensure: Accepted sequence $\mathbf{v}$ and next token y

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1: // Allocation Phase
2: Initialize  $\tilde{p}_r \leftarrow 1$ 
3: for each layer  $\ell = 0$  to  $D - 1$  do
4:   for each node  $v$  at layer  $\ell$  (in parallel) do
5:     Compute marginal probabilities  $p_v(u_i)$  for all
        $u_i \in \mathcal{C}(v)$  via Eqs. 4–6
6:     Compute rejection probability  $p_v(\neg v)$  (Eq. 7)
7:     // Conditional normalization for traversal order
8:      $m \leftarrow |\mathcal{C}(v)|$ 
9:     for  $j = 1$  to  $m$  do
10:       $\tilde{p}_{u_j} \leftarrow \frac{p_v(u_j)}{1 - \sum_{i=1}^{j-1} p_v(u_i)} \triangleright$  Conditional prob.
11:    end for
12:     $\tilde{p}_v^{\text{res}} \leftarrow 1 - \frac{p_v(\neg v)}{1 - \sum_{i=1}^m p_v(u_i)} \triangleright$  Fallback prob.
13:  end for
14: end for
15: // Decision Phase
16: Generate  $\eta \sim U(0, 1)^{|T|}$ 
17: for each node  $v$  in post-order traversal sequence do
18:   if  $v$  is leaf and  $\eta_v < \tilde{p}_v$  then
19:     Sample  $y \sim \mathcal{M}_b(\cdot | v)$  and return  $(v, y)$ 
20:   end if
21:   if  $v$  is non-leaf and  $\eta_v < \tilde{p}_v^{\text{res}}$  then
22:     Sample  $y \in \Sigma \setminus \mathcal{C}(v)$  from normalized residual
       distribution  $\frac{p_v(y)}{\sum_{u \notin \mathcal{C}(v)} p_v(u)}$  and return  $(v, y)$ 
23:   end if
24: end for
```

3.2 UniVer Algorithm

We now introduce the technical details of the UniVer pipeline (Algorithm 1), including its sampling strategy and computation of the core acceptance probability.

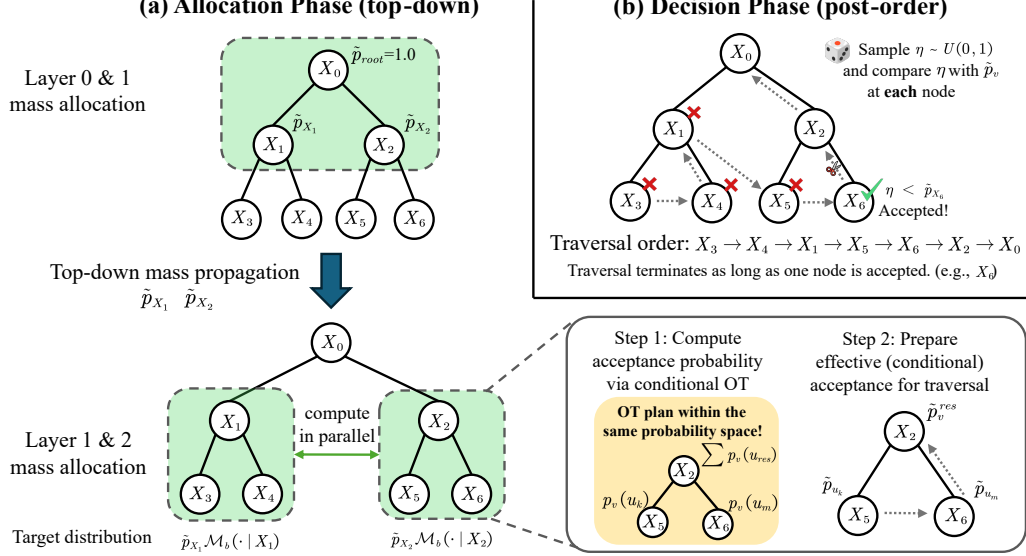


Figure 2: Two-stage verification framework of **UniVer**. **(a) Allocation Phase:** Acceptance probabilities are propagated in a top-down and layer-by-layer manner. For each branch within a layer, marginal probabilities are computed in parallel using conditional optimal transport, then normalized into conditional probabilities used for the next layer’s acceptance probability computation. **(b) Decision Phase:** A single-pass post-order traversal is executed. For each node X_i , a random variable η is compared against $\tilde{p}_{X_i}$ (for leaves) or $\tilde{p}_{X_i}^{\text{res}}$ (for non-leaves); the process terminates upon acceptance.

UniVer Acceptance Probability Computation. For any parent node v with effective conditional acceptance probability $\tilde{p}_v$ (precomputed recursively), its children nodes $\mathcal{C}(v) = \{u_1, \dots, u_m\}$ are selected by Greedy sampling strategy (i.e., Top_{m-1} plus $u_m \sim \mathcal{M}_s^-$).

For clarity, define the normalization factor:

$$Z_v = 1 - \tilde{p}_v + \sum_{x \in \Sigma} [\tilde{p}_v \mathcal{M}_b(x | v) - \mathcal{M}_s^-(x | v)]_+. \quad (3)$$

The marginal acceptance probabilities are given by:

$$p_v(u_m) = \min \left\{ 1, \frac{\tilde{p}_v \mathcal{M}_b(u_m | v)}{\mathcal{M}_s^-(u_m | v)} \right\} \quad (4)$$

$$p_v(u_k) = [\tilde{p}_v \mathcal{M}_b(u_k | v) - \mathcal{M}_s^-(u_k | v)]_+ \times \frac{1 - p_v(u_m)}{Z_v}, \quad \forall 1 \leq k \leq m-1, \quad (5)$$

$$p_v(u_{\text{res}}) = [\tilde{p}_v \mathcal{M}_b(u_{\text{res}} | v) - \mathcal{M}_s^-(u_{\text{res}} | v)]_+ \times \frac{1 - p_v(u_m)}{Z_v}, \quad \forall u_{\text{res}} \in \Sigma \setminus \mathcal{C}(v), \quad (6)$$

Then, the remain probability of rejecting v is

$$p_v(\neg v) = 1 - \sum_{u \in \Sigma} p_v(u) = \frac{(1 - p_v(u_m))(1 - \tilde{p}_v)}{Z_v}. \quad (7)$$

ensuring proper probability allocation.

4 Theoretical Guarantees

This section establishes the theoretical foundations of UniVer, proving three key properties:

- **Losslessness.** The algorithm preserves the target model distribution $\mathcal{M}_b$.
- **Conditional Optimality.** UniVer achieves the optimal acceptance rate for each single-layer subtree of T , given a prefix acceptance probability and its sampling strategy.
- **Superiority over Greedy.** The expected acceptance length of our UniVer is never worse than that of the Greedy method [8].

Preliminaries. We first formalize some key notations used in our analysis. Let p_{draft} be the probability distribution of generating a specific draft tree T under the draft model $\mathcal{M}_s$, satisfying $\sum_T p_{\text{draft}}(T) = 1$. For a verification algorithm $\mathcal{A}$, we write $\mathcal{A} = o'$ to indicate that the algorithm outputs exactly the sequence o' . Note that o' except the last one form a valid root-to-node path in T , while its last element is a sample token y . Each *prefix acceptance probability* $\tilde{p}_v \in [0, 1]$ has been evaluated recursively by Line 10 of Algorithm 1.

Detailed proofs of the following theoretical results are referred to Appendix B.

Definition 1 (Locally Lossless Tree). *Let T be a draft tree equipped with acceptance probabilities $\{\tilde{p}_v\}_{v \in T}$ and $\{p_v(u)\}_{u \in \mathcal{C}(v)}$ generated by the Allocation Phase. If for any non-leaf node $v \in T$, the following property holds:*

$$\mathbb{E}_{\mathcal{C}(v)}[p_v(t)] = \tilde{p}_v \cdot \mathcal{M}_b(t | v), \quad \forall t \in \Sigma, \quad (8)$$

where $\mathbb{E}_{\mathcal{C}(v)}[\cdot]$ denotes expectation over the randomness of generating the children set $\mathcal{C}(v)$, then T is called a **locally lossless tree**.

Definition 2 (Lossless Verification). *A verification algorithm $\mathcal{A}(T, \mathcal{M}_s, \mathcal{M}_b)$ is **lossless** if for any sequence $o = (o_0, o_1, \dots, o_L) \in \Sigma^{L+1}$ with $o_0 = r$ being the root, the following equation holds:*

$$\sum_T p_{\text{draft}}(T) \sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' | T]}{\mathcal{M}_b(o')} = \mathcal{M}_b(o), \quad (9)$$

where the inner summation ranges over all prefix sequences $o' \subseteq o$ (including $o' = o$ itself), and $\frac{\mathcal{M}_b(o)}{\mathcal{M}_b(o')}$ is the conditional probability of generating the remaining tokens $\{o \setminus o'\}$ by $\mathcal{M}_b(\cdot | o')$.

Theorem 3 (Local losslessness of UniVer). *The draft tree equipped with the acceptance probabilities generated in Allocation Phase of UniVer verification is a locally lossless tree.*

Theorem 4 (General Losslessness). *For any **two-stage** verification algorithm $\mathcal{A}(T, \mathcal{M}_s, \mathcal{M}_b)$ consisting of Allocation and Decision Phases, if the draft tree generated in the Allocation Phase is **locally lossless**, then the verification $\mathcal{A}$ is **lossless**.*

Corollary 1 (Losslessness of UniVer). *From Theorems 3 and 4, UniVer produces tokens are consistent with the exact target distribution $\mathcal{M}_b$, i.e., UniVer is **strictly lossless**.*

Theorem 5 (Conditional Optimality). *Let T be any locally lossless tree satisfying Definition 1 and let $v \in T$ be a non-leaf node with effective acceptance probability $\tilde{p}_v$. Then UniVer achieves the maximum conditional acceptance rate for the children $\mathcal{C}(v) = \{u_1, \dots, u_m\}$ (over the randomness of $u_m \sim \mathcal{M}_s^-(\cdot | v)$), which equals:*

$$\begin{aligned} & \alpha_{\text{UniVer}}^*(\tilde{p}_v \mathcal{M}_b(\cdot | v), \mathcal{M}_s(\cdot | v)) \\ &= \sum_{u \in \text{Top}_{m-1}(\mathcal{M}_s(\cdot | v))} \tilde{p}_v \mathcal{M}_b(u | v) + \sum_{u \in \Sigma} \min\{\tilde{p}_v \mathcal{M}_b(u | v), \mathcal{M}_s^-(u | v)\}. \end{aligned} \quad (10)$$

Theorem 6. *For any draft tree $T \sim p_{\text{draft}}$, let $N_{\text{UniVer}}(T)$ and $N_{\text{Greedy}}(T)$ denote the acceptance lengths of UniVer and the Greedy method [8], respectively. Then UniVer exhibits the following superiority:*

$$\mathbb{E}_{T \sim p_{\text{draft}}}[N_{\text{UniVer}}(T)] \geq \mathbb{E}_{T \sim p_{\text{draft}}}[N_{\text{Greedy}}(T)]. \quad (11)$$

5 Experiments

5.1 Experimental Setup

Datasets. We conduct the experiments on Spec-Bench [26], following the setup of Traversal Verification [24]. Spec-Bench encompasses six distinct domains, each with 80 representative samples: multi-turn conversation on MT-Bench [29], translation on WMT14 DE-EN [1], summarization on CNN/Daily Mail [18], question answering on Natural Questions [12], mathematical reasoning on GSM8K [5], and retrieval-augmented generation on DPR [10].

Models. We mainly conduct experiments on the Vicuna [29] model family, using EAGLE [15] as the draft model. This pairing is widely adopted in previous research [19, 24, 8]. We also include experiments on other models such as Llama3.1-8B-Instruct [6] to validate its generalization. Due to space limits, we do not include all results in the main text. Please refer to Appendix A for comprehensive experiments with different drafter structures and model sizes.

Table 1: Acceptance length (τ) and throughput on Vicuna-7B-v1.3. Autoregressive throughput baseline: 35.9 tokens/s. Best results are highlighted in **bold**.

Method		MT	Trans.	Summ.	QA	Math	RAG	Avg. τ (Δ)	Avg. TPS
RRSw-based	RRSw	3.42 $\pm$ 0.03	2.73 $\pm$ 0.03	2.97 $\pm$ 0.02	2.72 $\pm$ 0.01	3.47 $\pm$ 0.06	3.05 $\pm$ 0.01	3.06 $\pm$ 0.02 (0.0%)	67.5 $\pm$ 0.2
	Traversal	3.49 $\pm$ 0.03	2.75 $\pm$ 0.04	3.03 $\pm$ 0.02	2.79 $\pm$ 0.03	3.55 $\pm$ 0.06	3.15 $\pm$ 0.06	3.13 $\pm$ 0.02 (2.3% $\uparrow$)	68.9 $\pm$ 0.5
OT-based	Greedy	3.56 $\pm$ 0.04	2.80 $\pm$ 0.03	3.12 $\pm$ 0.03	2.87 $\pm$ 0.05	3.66 $\pm$ 0.06	3.24 $\pm$ 0.03	3.21 $\pm$ 0.02 (4.9% $\uparrow$)	70.4 $\pm$ 0.6
	UniVer	3.68$\pm$0.02	2.83$\pm$0.04	3.21$\pm$0.04	2.94$\pm$0.02	3.76$\pm$0.02	3.32$\pm$0.05	3.29$\pm$0.01 (7.5% $\uparrow$)	72.2$\pm$0.5

Table 2: Acceptance length (τ) and throughput on Llama3.1-8B-Instruct. Autoregressive throughput baseline: 34.5 tokens/s. Best results are highlighted in **bold**.

Method		MT	Trans.	Summ.	QA	Math	RAG	Avg. τ (Δ)	Avg. TPS
RRSw-based	RRSw	2.96 $\pm$ 0.03	2.31 $\pm$ 0.03	2.56 $\pm$ 0.01	2.33 $\pm$ 0.01	3.31 $\pm$ 0.06	2.86 $\pm$ 0.04	2.72 $\pm$ 0.02 (0.0%)	48.5 $\pm$ 0.2
	Traversal	3.05 $\pm$ 0.02	2.34 $\pm$ 0.06	2.60 $\pm$ 0.01	2.40 $\pm$ 0.05	3.37 $\pm$ 0.05	2.94 $\pm$ 0.04	2.78 $\pm$ 0.02 (2.2% $\uparrow$)	49.5 $\pm$ 0.4
OT-based	Greedy	3.09 $\pm$ 0.01	2.54 $\pm$ 0.04	2.70 $\pm$ 0.02	2.46 $\pm$ 0.02	3.45 $\pm$ 0.03	3.04 $\pm$ 0.06	2.88 $\pm$ 0.02 (5.9% $\uparrow$)	50.6 $\pm$ 0.4
	UniVer	3.19$\pm$0.02	2.57$\pm$0.06	2.77$\pm$0.01	2.57$\pm$0.04	3.51$\pm$0.03	3.11$\pm$0.04	2.95$\pm$0.01 (8.5% $\uparrow$)	52.2$\pm$0.1

Implementation and Comparisons. We implement UniVer and baseline methods based on the EAGLE [15] open-source repository. All experiments are conducted on NVIDIA RTX A6000 48G GPUs, with each configuration running on three different random seeds. We report the *acceptance length* τ (average tokens generated per cycle) by mean $\pm$ std as the primary metric, which reflects the theoretical efficiency of the verification algorithm. We also measure the practical throughput (Tokens Per Second, TPS) for a comprehensive comparison. We compare UniVer against existing verification methods spanning two paradigms:

1. **RRSw-based:** (1) *Vanilla RRSw*: Standard recursive rejection sampling without replacement applied independently at each node, as implemented in EAGLE [15]; (2) *Traversal*: The Traversal Verification algorithm [24] which extends Block Verification to trees using RRSw for horizontal draft selection.
2. **OT-based:** (1) *Greedy method* [8]: Single-step OT with Greedy draft selection; (2) *UniVer* (Ours): Our proposed unified framework that extends OT-based verification to tree structures.

5.2 Overall Performance

Table 1 presents the results on Vicuna-7B-v1.3 using a balanced binary tree of depth 5 (32 leaf nodes) under temperature 1.0. UniVer achieves a τ of 3.29, representing a 7.5% improvement over the Vanilla RRSw baseline. Notably, Traversal Verification, which extends Block Verification to trees using RRSw, only marginally outperforms Vanilla RRSw (2.3%), suggesting that bottom-up traversal alone provides limited benefit when horizontal draft selection remains suboptimal. In contrast, UniVer achieves a 7.5% higher τ over the vanilla baseline and a 2.5% gain over Greedy method [8].

Figure 3 demonstrates the per-depth acceptance rates on MT-bench. At depth 0, Greedy and UniVer both achieve 82.6%, outperforming RRSw-based methods through coordinated horizontal selection. All methods exhibit a characteristic drop at depth 1, consistent with the known degradation of EAGLE draft model alignment beyond the first token [28, 25]. As depth increases, RRSw degrades steadily while Traversal stabilizes around 75% via vertical probability recharging but plateaus. Greedy enjoys strong early-layer acceptance, but shares similar decay patterns like RRSw. UniVer maintains acceptance above 77% in all depths, confirming that it successfully propagates the probability mass across depths and benefits from horizontal-vertical joint optimization.

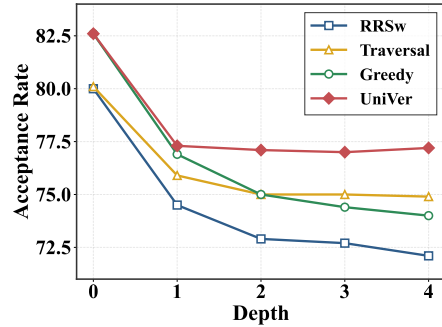


Figure 3: Per-depth acceptance rates on Vicuna-7B-v1.3 (binary tree, depth 5).

In terms of wall-clock efficiency, since UniVer only uses one sampled token with other tokens selected deterministically, the acceptance probabilities of all candidates admit closed-form expressions without using OT solvers. The overall latency of UniVer is therefore comparable to that of standard RRSw, and the acceptance gains directly translate into throughput improvements. To contextualize these numbers: for Vicuna-7B-v1.3, while UniVer improves acceptance length by 7.5%, the end-to-end throughput gain is approximately 7%.

Similar trends are observed on Llama3.1-8B-Instruct (Table 2), confirming its robustness across different architectures. For full experimental results, please refer to Appendix A.

5.3 Scaling with Tree Size

As shown in Figure 4, the advantage of UniVer becomes more pronounced as the tree size and depth increase. Compared to single-step OT (Greedy method), UniVer achieves larger gains when the tree gets deeper. This trend arises because UniVer’s conditional OT framework effectively propagates probability mass through hierarchical dependencies, whereas Greedy applies independent single-step optimization at each node, as the local truncation of probabilities compounds across hierarchical levels. Similarly, as depth grows, Traversal demonstrates increasing advantages over Vanilla RRSw, validating that sequence-level verification benefits from deeper trees. However, UniVer consistently outperforms Traversal across all depths, as conditional OT achieves superior horizontal draft selection compared to RRSw’s local pairwise comparisons.

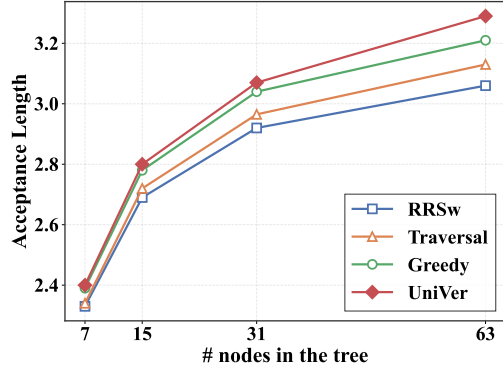


Figure 4: Acceptance length as a function of binary tree size (number of nodes) on Vicuna-7B-v1.3. Tree depth ranges from 2 to 5, corresponding to 7, 15, 31, and 63 nodes respectively. The temperature is set to 1.0.

5.4 Effect of Tree Topology and Temperature

Table 3 examines how UniVer performs across varying tree architectures and sampling temperatures.

Impact of Temperature. As the temperature decreases, the output distributions become more concentrated, causing the performance gap between verification methods to narrow. At temperature 0.3, Traversal, Greedy, and UniVer achieve nearly identical acceptance lengths (e.g., 3.39 for the binary tree with depth = 5 configuration), converging toward the same outcome as observed at temperature 0 (3.40), where the deterministic nature renders these verification methods equivalent. Conversely, at higher temperatures, where distributions are more dispersed, UniVer demonstrates more pronounced advantages over existing methods.

Impact of Tree Topology. UniVer maintains consistent improvements across diverse tree structures, including balanced trees (2 drafts $\times$ 5 steps and 3 drafts $\times$ 3 steps) and the imbalanced EAGLE tree. While the absolute gain varies with topology (e.g., 7.5% on the balanced 2x5 tree versus 4.2% on the EAGLE tree at temperature 1.0), UniVer consistently achieves the highest acceptance lengths. This variation arises because the optimal transport formulation allocates probability mass globally across siblings, whereas the relative benefit of such global coordination depends on the specific branching structure and depth distribution of the tree.

6 Related Work

Speculative Decoding. Introduced by Leviathan et al. [13] and Chen et al. [3], speculative decoding accelerates LLM inference via a draft-then-verify paradigm. Research in this area primarily focuses on two directions: (1) improving draft quality and tree structures, including architectural innovations such as EAGLE series [15, 16], Medusa [2] and tree topology design, like EAGLE-2 [14] and Sequoia [4]; (2) designing verification algorithms, which is most relevant to this work.

Table 3: Impact of temperature and tree topology on acceptance length (τ) using Vicuna-7B-v1.3. Results show mean $\pm$ std. Δ indicates relative improvement over Vanilla RRSw.

Method		Drafts=2 Steps=5		Drafts=3 Steps=3		EAGLE Tree	
		Avg. τ	Δ	Avg. τ	Δ	Avg. τ	Δ
Temperature = 0.3							
RRSw-based	Vanilla RRSw	3.34 $\pm$ 0.01	0.0%	3.04 $\pm$ 0.01	0.0%	3.33 $\pm$ 0.01	0.0%
	Traversal	3.39$\pm$0.01	1.5%	3.08 $\pm$ 0.01	1.3%	3.38$\pm$0.01	1.5%
OT-based	Greedy	3.39$\pm$0.01	1.5%	3.08 $\pm$ 0.01	1.3%	3.38$\pm$0.01	1.5%
	UniVer	3.39$\pm$0.01	1.5%	3.09$\pm$0.01	1.6%	3.38$\pm$0.01	1.5%
Temperature = 0.6							
RRSw-based	Vanilla RRSw	3.29 $\pm$ 0.01	0.0%	3.02 $\pm$ 0.01	0.0%	3.29 $\pm$ 0.01	0.0%
	Traversal	3.32 $\pm$ 0.01	0.9%	3.05 $\pm$ 0.01	1.0%	3.33 $\pm$ 0.01	1.2%
OT-based	Greedy	3.34 $\pm$ 0.02	1.5%	3.07 $\pm$ 0.01	1.7%	3.34 $\pm$ 0.01	1.5%
	UniVer	3.37$\pm$0.01	2.4%	3.09$\pm$0.01	2.3%	3.35$\pm$0.01	1.8%
Temperature = 1.0							
RRSw-based	Vanilla RRSw	3.06 $\pm$ 0.01	0.0%	2.89 $\pm$ 0.01	0.0%	3.11 $\pm$ 0.01	0.0%
	Traversal	3.13 $\pm$ 0.02	2.3%	2.91 $\pm$ 0.01	0.7%	3.16 $\pm$ 0.01	1.6%
OT-based	Greedy	3.21 $\pm$ 0.02	4.9%	3.00 $\pm$ 0.01	3.8%	3.21 $\pm$ 0.01	3.2%
	UniVer	3.29$\pm$0.01	7.5%	3.03$\pm$0.01	4.8%	3.24$\pm$0.02	4.2%
Reference (Temperature = 0)		3.40	-	3.09	-	3.40	-

Multi-Step Verification. Methods in this direction extend speculative decoding beyond single-step generation to handle sequential dependencies. For single-chain decoding, Block Verification [20] and ASpS [7] identify the sub-optimality of local (per-step) verification and propose sequence-level alternatives that maximize expected acceptance length along the chain. Weng et al. [24] further extend this to tree structures through Traversal Verification, which processes nodes bottom-up.

Multi-Draft Verification. Complementary to multi-step methods, this direction focuses on optimizing the acceptance rate for multiple candidates generated at a single step. SpecTr [21] formulates verification as an optimal transport problem and has been further improved by subsequent works [11, 22]. SpecInfer [17] employs Recursive Rejection Sampling (RRS), later refined to RRSw [9, 27, 4] to prevent redundant sampling of identical tokens. Beyond standard sampling strategies, SpecHub [19] and Greedy method [8] propose hybrid approaches that deterministically select high-probability candidates with stochastic sampled tokens, with the latter achieving the theoretical optimal acceptance rate under its drafting strategy within single-step.

7 Conclusion

In this work, we addressed the fragmentation in speculative decoding verification, where existing methods treat multi-draft and multi-step optimizations in isolation. We revealed that longitudinal dependencies can be formulated as dynamic scaling factors via prefix acceptance probabilities and framed tree-based verification as a conditional OT problem.

Building on this theoretical foundation, we proposed UniVer, an algorithm that composes local optimal transport plans under prefix constraints to jointly optimize across tree levels. We proved that UniVer achieves the optimal acceptance rate within this conditional framework while maintaining strict losslessness. Empirical results demonstrate that UniVer significantly improves acceptance length over existing methods, validating the efficacy of our unified verification strategy.

Limitations and Future Work. Distinct verification methods only exhibit performance differences when the sampling temperature is above zero. Therefore, UniVer possesses no extra performance gain under or near temperature = 0. Beyond this universal boundary, the realized acceptance length is intrinsically linked to the interplay between tree topology and sampling strategy. UniVer relies on specific mixed sampling strategy to derive OT-based verification, while co-designing the sampling strategy, verification mechanism, and the tree structure to maximize longitudinal mass propagation remains a promising direction for future research.

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A Additional Experimental Results

We provide additional experimental results on Vicuna-13B-v1.3, Vicuna-33B-v1.3 and Qwen2-7B-Instruct with their corresponding EAGLE draft model in Table 4, Table 5, and Table 6. We also include experiments on Llama2-7B [23] with Llama-68M [17] as the draft model in Table 7.

Table 4: Acceptance length (τ) and throughput on Vicuna-13B-v1.3. Autoregressive throughput baseline: 22.3 tokens/s. Best results are highlighted in **bold**.

Method		MT	Trans.	Summ.	QA	Math	RAG	Avg. τ (Δ)	Avg. TPS
RRSw-based	RRSw	3.57 $\pm$ 0.06	2.81 $\pm$ 0.03	3.17 $\pm$ 0.03	2.77 $\pm$ 0.05	3.63 $\pm$ 0.08	3.11 $\pm$ 0.05	3.17 $\pm$ 0.02 (0.0%)	44.3 $\pm$ 0.2
	Traversal	3.60 $\pm$ 0.05	2.82 $\pm$ 0.02	3.19 $\pm$ 0.03	2.85 $\pm$ 0.05	3.71 $\pm$ 0.05	3.15 $\pm$ 0.04	3.22 $\pm$ 0.01 (1.6% $\uparrow$)	44.9 $\pm$ 0.1
OT-based	Greedy	3.67 $\pm$ 0.02	2.92 $\pm$ 0.04	3.27 $\pm$ 0.01	2.91 $\pm$ 0.04	3.79 $\pm$ 0.03	3.28 $\pm$ 0.04	3.30 $\pm$ 0.02 (4.1% $\uparrow$)	45.7 $\pm$ 0.3
	UniVer	3.74$\pm$0.02	2.97$\pm$0.03	3.31$\pm$0.03	2.97$\pm$0.01	3.84$\pm$0.10	3.30$\pm$0.03	3.35$\pm$0.01 (5.7% $\uparrow$)	46.5$\pm$0.4

Table 5: Acceptance length (τ) and throughput on Vicuna-33B-v1.3. Autoregressive throughput baseline: 9.2 tokens/s. Best results are highlighted in **bold**.

Method		MT	Trans.	Summ.	QA	Math	RAG	Avg. τ (Δ)	Avg. TPS
RRSw-based	RRSw	3.38 $\pm$ 0.04	2.68 $\pm$ 0.01	2.97 $\pm$ 0.01	2.66 $\pm$ 0.02	3.70 $\pm$ 0.01	2.94 $\pm$ 0.02	3.05 $\pm$ 0.02 (0.0%)	20.6 $\pm$ 0.1
	Traversal	3.41 $\pm$ 0.01	2.72 $\pm$ 0.01	3.01 $\pm$ 0.01	2.68 $\pm$ 0.04	3.73 $\pm$ 0.04	2.98 $\pm$ 0.03	3.09 $\pm$ 0.01 (1.1% $\uparrow$)	20.8 $\pm$ 0.1
OT-based	Greedy	3.50 $\pm$ 0.01	2.80 $\pm$ 0.02	3.07 $\pm$ 0.01	2.78 $\pm$ 0.01	3.82 $\pm$ 0.03	3.05 $\pm$ 0.03	3.17 $\pm$ 0.01 (3.8% $\uparrow$)	21.2 $\pm$ 0.1
	UniVer	3.52$\pm$0.01	2.83$\pm$0.02	3.11$\pm$0.03	2.81$\pm$0.06	3.86$\pm$0.01	3.10$\pm$0.05	3.21$\pm$0.01 (4.9% $\uparrow$)	21.4$\pm$0.1

Table 6: Acceptance length (τ) and throughput on Qwen2-7B-Instruct. Autoregressive throughput baseline: 37.0 tokens/s. Best results are highlighted in **bold**.

Method		MT	Trans.	Summ.	QA	Math	RAG	Avg. τ (Δ)	Avg. TPS
RRSw-based	RRSw	2.09 $\pm$ 0.01	2.57 $\pm$ 0.03	1.44 $\pm$ 0.01	2.37 $\pm$ 0.03	3.09 $\pm$ 0.02	1.48 $\pm$ 0.02	2.17 $\pm$ 0.01 (0.0%)	43.7 $\pm$ 0.3
	Traversal	2.14 $\pm$ 0.01	2.61 $\pm$ 0.03	1.45 $\pm$ 0.01	2.43 $\pm$ 0.04	3.12 $\pm$ 0.03	1.50 $\pm$ 0.02	2.21 $\pm$ 0.01 (1.8% $\uparrow$)	44.0 $\pm$ 0.3
OT-based	Greedy	2.15 $\pm$ 0.02	2.64 $\pm$ 0.04	1.49 $\pm$ 0.02	2.46 $\pm$ 0.05	3.21 $\pm$ 0.01	1.53 $\pm$ 0.02	2.25 $\pm$ 0.02 (3.7% $\uparrow$)	44.7 $\pm$ 0.3
	UniVer	2.20$\pm$0.01	2.70$\pm$0.04	1.50$\pm$0.01	2.57$\pm$0.04	3.26$\pm$0.02	1.56$\pm$0.02	2.30$\pm$0.01 (6.0% $\uparrow$)	45.8$\pm$0.2

Table 7: Acceptance length (τ) and throughput on Llama2-7B. Autoregressive throughput baseline: 37.3 tokens/s. Best results are highlighted in **bold**.

Method		MT	Trans.	Summ.	QA	Math	RAG	Avg. τ (Δ)	Avg. TPS
RRSw-based	RRSw	2.47 $\pm$ 0.03	2.41 $\pm$ 0.04	2.15 $\pm$ 0.03	2.60 $\pm$ 0.02	2.45 $\pm$ 0.06	2.52 $\pm$ 0.11	2.43 $\pm$ 0.04 (0.0%)	57.9 $\pm$ 1.3
	Traversal	2.55 $\pm$ 0.04	2.49 $\pm$ 0.09	2.22 $\pm$ 0.04	2.65 $\pm$ 0.06	2.58 $\pm$ 0.06	2.69 $\pm$ 0.02	2.53 $\pm$ 0.03 (4.1% $\uparrow$)	59.6 $\pm$ 1.1
OT-based	Greedy	2.48 $\pm$ 0.04	2.52 $\pm$ 0.07	2.27 $\pm$ 0.03	2.68 $\pm$ 0.04	2.53 $\pm$ 0.06	2.71 $\pm$ 0.04	2.53 $\pm$ 0.03 (4.1% $\uparrow$)	59.1 $\pm$ 1.0
	UniVer	2.73$\pm$0.01	2.59$\pm$0.09	2.31$\pm$0.04	2.84$\pm$0.04	2.66$\pm$0.04	2.77$\pm$0.07	2.65$\pm$0.03 (9.1% $\uparrow$)	62.5$\pm$0.4

B The Proof of Theorems

Vocabulary and Sequences. Let Σ denote a finite vocabulary. For any positive integer L , we use Σ^L denote the set of all sequences (strings) of length L over Σ . We write $a = (a_0, \dots, a_L) \in \Sigma^{L+1}$ to represent a sequence of length $|a| = L + 1$, and $a_{[:\ell]}$ to represent the prefix $(a_0, \dots, a_\ell)$ consisting of the first ℓ elements of a .

B.1 Local losslessness of UniVer

Proof of Theorem 3. Considering the draft tree T generated by *Allocation Phase* of UniVer verification (see Section 3.2), we observe that, for any fixed non-leaf node $v \in T$, the randomness of its children set $\mathcal{C}(v) = \{u_1, \dots, u_m\}$ only depends on the last sampling token $u_m \sim \mathcal{M}_s^\top(\cdot \mid v)$, which means $\Pr(\text{draft } \mathcal{C}(v)) = \mathcal{M}_s^\top(u_m \mid v)$.

Now we begin to prove $\mathbb{E}_{\mathcal{C}(v)}[p_v(t)] = \tilde{p}_v \mathcal{M}_b(t \mid v)$ holds for all $t \in \Sigma$. First, observe that the normalization factor Z_v in Equations (3) is actually independent of the sampled value u_m . We use

$H_v = \{u_1, \dots, u_{m-1}\}$ represent the top- $(m-1)$ tokens, and then

$$\begin{aligned}
Z_v &= 1 - \tilde{p}_v + \sum_{x \in \Sigma} [\tilde{p}_v \mathcal{M}_b(x|v) - \mathcal{M}_s^-(x|v)]_+ \\
&= 1 - \tilde{p}_v + \sum_{x \in H_v} \tilde{p}_v \mathcal{M}_b(x|v) + \sum_{x \notin H_v} [\tilde{p}_v \mathcal{M}_b(x|v) - \mathcal{M}_s^-(x|v)]_+ \\
&= 1 - \tilde{p}_v \sum_{x \notin H_v} \mathcal{M}_b(x|v) + \sum_{x \notin H_v} [\tilde{p}_v \mathcal{M}_b(x|v) - \mathcal{M}_s^-(x|v)]_+ \\
&= 1 - \sum_{x \notin H_v} \min\{\tilde{p}_v \mathcal{M}_b(x|v), \mathcal{M}_s^-(x|v)\}.
\end{aligned} \tag{12}$$

For any $t \in \Sigma$, we consider the following two cases.

Case 1: $t \in H_v$ (deterministic nodes). From Equation (5) and noting that $\mathcal{M}_s^-(t | v) = 0$ for $t \in H_v$, we have

$$\mathbb{E}_{\mathcal{C}(v)}[p_v(t)] = \frac{\tilde{p}_v \mathcal{M}_b(t | v)}{Z_v} \cdot \mathbb{E}_{u_m}[1 - p_v(u_m)].$$

By Equation (4), we know

$$\mathbb{E}_{u_m}[1 - p_v(u_m)] = 1 - \sum_{x \notin H_v} \min\{\mathcal{M}_s^-(x|v), \tilde{p}_v \mathcal{M}_b(x|v)\}.$$

The expectation term equals Z_v (see Equation (12)). Thus $\mathbb{E}_{u_m}[p_v(t)] = \tilde{p}_v \mathcal{M}_b(t | v)$.

Case 2: $t \notin H_v$ (residual vocabulary). Here t may either be the sampled node u_m or belong to the remaining residual set. Obviously,

$$\begin{aligned}
\mathbb{E}_{u_m}[p_v(t)] &= \mathcal{M}_s^-(t | v) \cdot \min\left\{1, \frac{\tilde{p}_v \mathcal{M}_b(t | v)}{\mathcal{M}_s^-(t | v)}\right\} \\
&+ \frac{[\tilde{p}_v \mathcal{M}_b(t | v) - \mathcal{M}_s^-(t | v)]_+}{Z_v} \times \sum_{x \notin H_v \cup \{t\}} \mathcal{M}_s^-(x | v) \cdot (1 - p_v(x)).
\end{aligned}$$

Since

$$\begin{aligned}
&\sum_{x \notin H_v \cup \{t\}} \mathcal{M}_s^-(x | v) (1 - p_v(x)) \\
&= \sum_{x \notin H_v \cup \{t\}} \mathcal{M}_s^-(x | v) - \sum_{x \notin H_v \cup \{t\}} \min\{\mathcal{M}_s^-(x | v), \tilde{p}_v \mathcal{M}_b(x | v)\} \\
&= Z_v + \min\{0, \tilde{p}_v \mathcal{M}_b(t | v) - \mathcal{M}_s^-(t | v)\}.
\end{aligned}$$

Substituting back, we obtain

$$\begin{aligned}
\mathbb{E}_{u_m}[p_v(t)] &= \min\{\mathcal{M}_s^-(t | v), \tilde{p}_v \mathcal{M}_b(t | v)\} + [\tilde{p}_v \mathcal{M}_b(t | v) - \mathcal{M}_s^-(t | v)]_+ \\
&= \tilde{p}_v \mathcal{M}_b(t | v).
\end{aligned}$$

Therefore, the draft tree generated in UniVer verification is a locally lossless tree. $\square$

B.2 Losslessness of two-stage verification

Proof of Theorem 4. We prove it by induction on the number of parent nodes n in tree T .

Base case ($n = 1$): The tree T consists of only the root node r and its children $\mathcal{C}(r) = \{u_1, \dots, u_m\}$. The depth of T is 1 (root at depth 0, children at depth 1), so we consider any output sequence $o = (r, w, y) \in \Sigma^3$ where $w, y \in \Sigma$.

In the Decision Phase, the traversal order is $(u_1, \dots, u_m, r)$. There are two cases for how the algorithm $\mathcal{A}$ can generate o :

Case 1: $w \in \mathcal{C}(r)$. The algorithm $\mathcal{A}$ accepts the child node $w = u_i$ for some $1 \leq i \leq m$ and then samples $y \sim \mathcal{M}_b(\cdot | r, w)$. The probability of accepting node w is:

$$\tilde{p}_{u_i} \prod_{j < i} (1 - \tilde{p}_{u_j}) = p_r(w),$$

where $p_r(w)$ is the marginal acceptance mass computed in the Allocation Phase. So the probability of $\mathcal{A}$ outputs o equals

$$\Pr[\mathcal{A} = o \mid w \in \mathcal{C}(r)] = p_r(w) \mathcal{M}_b(y \mid r, w).$$

Then the contribution to the left-hand side of Equation (9) is

$$\begin{aligned} & \sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' \mid T]}{\mathcal{M}_b(o')} \\ &= \Pr[\mathcal{A} = o \mid w \in \mathcal{C}(r)] \\ &= p_r(w) \mathcal{M}_b(y \mid r, w), \quad (\text{when } w \in \mathcal{C}(r)) \end{aligned}$$

Case 2: $w \notin \mathcal{C}(r)$. The algorithm $\mathcal{A}$ rejects all children and accepts the root node r , then samples w from the residual distribution $\mathcal{M}_b(\cdot \mid r)$. The probability of accepting the root (fallback) is:

$$\prod_{j=1}^m (1 - \tilde{p}_{u_j}) = 1 - \sum_{u \in \mathcal{C}(r)} p_r(u) = \sum_{u \notin \mathcal{C}(r)} p_r(u).$$

So the probability of $\mathcal{A}$ outputs $o' = (r, w)$ equals

$$\Pr[\mathcal{A} = (r, w) \mid w \notin \mathcal{C}(r)] = \left(\sum_{u \notin \mathcal{C}(r)} p_r(u) \right) \frac{p_r(w)}{\sum_{u \notin \mathcal{C}(r)} p_r(u)} = p_r(w).$$

Then the contribution to the left-hand side of Equation (9) is:

$$\begin{aligned} & \sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' \mid T]}{\mathcal{M}_b(o')} \\ &= \mathcal{M}_b(y \mid r, w) \Pr[\mathcal{A} = (r, w) \mid w \notin \mathcal{C}(r)] \\ &= p_r(w) \mathcal{M}_b(y \mid r, w), \quad (\text{when } w \notin \mathcal{C}(r)) \end{aligned}$$

Combining both cases, for any sequence $o = (r, w, y)$, we have:

$$\begin{aligned} & \sum_T p_{\text{draft}}(T) \sum_{o' \subseteq o} \frac{\mathcal{M}_b(o)}{\mathcal{M}_b(o')} \Pr[\mathcal{A} = o' \mid T] \\ &= \sum_T p_{\text{draft}}(T) \cdot p_r(w) \mathcal{M}_b(y \mid r, w) \\ &= \mathcal{M}_b(y \mid r, w) \mathbb{E}_{\mathcal{C}(r)}[p_r(w)]. \end{aligned}$$

By the locally lossless property (Definition 1), for any specific token $w \in \Sigma$, we have $\mathbb{E}_{\mathcal{C}(r)}[p_r(w)] = \tilde{p}_r \mathcal{M}_b(w \mid r) = \mathcal{M}_b(w \mid r)$. Thus, the base case $n = 1$ holds.

Inductive step. Assume the theorem holds for all locally lossless trees with at most K parent nodes. Consider a locally lossless tree T with $K + 1$ parent nodes. Let v be a deepest parent node in T , which implies all children $\mathcal{C}(v)$ are leaves. Construct the reduced tree $T^{\text{sub}} = T \setminus \mathcal{C}(v)$ by removing all children of v . Then T^{sub} contains exactly K parent nodes and satisfies the induction hypothesis, that is for any sequence $o \in \Sigma^{L+1}$,

$$\mathbb{E}_{T^{\text{sub}}} \left[\sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' \mid T^{\text{sub}}]}{\mathcal{M}_b(o')} \right] = \mathcal{M}_b(o). \quad (13)$$

For any sequence $o' = (o_0, \dots, o_\ell)$, $\ell \leq L$, as the output of $\mathcal{A}(T, \mathcal{M}_s, \mathcal{M}_b)$. Then $o_{\ell-1}$ is the acceptance node in Decision Phase. We divide into three cases to prove the following equation

$$\mathbb{E}_{\mathcal{C}(v)} \left[\sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' \mid T]}{\mathcal{M}_b(o')} \right] = \sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' \mid T^{\text{sub}}]}{\mathcal{M}_b(o')}. \quad (14)$$

Case 1: the acceptance node $o_{\ell-1}$ is a preceding node of $\mathcal{C}(v)$. In this case, we obviously have

$$\Pr[\mathcal{A} = o' \mid T] = \Pr[\mathcal{A} = o' \mid T^{\text{sub}}]. \quad \forall o'$$

Then (14) always holds for this case.

Case 2: the acceptance node $o_{\ell-1} \in \mathcal{C}(v) \cup \{v\}$. Since Case 1 occurs with the same probability for both the trees T and T^{sub} , we only need to consider the conditional probability under the condition that Case 1 does not occur.

If $o_{\ell-1} \in \mathcal{C}(v)$ (i.e., $o' = o$ and $\ell = L$), then the probability

$$\sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' \mid T]}{\mathcal{M}_b(o')} = p_v(o_{L-1}) \mathcal{M}_b(o_L \mid o_{[:L-1]}), \quad (o_{L-1} \in \mathcal{C}(v)).$$

If $o_{\ell-1} = v$ (i.e., $o' = o_{[:L-1]}$ and $\ell = L - 1$), then the probability

$$\begin{aligned} & \sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' \mid T]}{\mathcal{M}_b(o')} \\ &= \Pr[\mathcal{A} = o' \mid T] \mathcal{M}_b(o_L \mid o') \\ &= p_v(o_{L-1}) \mathcal{M}_b(o_L \mid o_{[:L-1]}), \quad (o_{L-1} \notin \mathcal{C}(v)). \end{aligned}$$

To sum up, by the locally lossless of T , we know the Equation (14)

$$\begin{aligned} & \mathbb{E}_{\mathcal{C}(v)} \left[\sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' \mid T]}{\mathcal{M}_b(o')} \right] \\ &= \mathcal{M}_b(o_L \mid o_{[:L-1]}) \cdot \mathbb{E}_{\mathcal{C}(v)}[p_v(o_{L-1})] \\ &\stackrel{(8)}{=} \mathcal{M}_b(o_L \mid o_{[:L-1]}) \cdot \tilde{p}_v \mathcal{M}_b(o_L \mid v, o_{L-1}) \\ &= \mathcal{M}_b(o_L \mid o_{[:L-1]}) \Pr[\mathcal{A} = (v, o_{L-1}) \mid T^{\text{sub}}] \\ &= \sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' \mid T^{\text{sub}}]}{\mathcal{M}_b(o')} \end{aligned}$$

holds for this case.

Case 3: the acceptance node $o_{\ell-1}$ is an after-node of v . Since after rejection at node v , the remaining parts of T and T^{sub} are identical under the Decision Phase. Therefore, we only need to prove that the expectation of Case 3 occurring for T equals to the probability for T^{sub} rejecting v , i.e., to show that

$$\mathbb{E}_{\mathcal{C}(v)} \Pr(\mathcal{A} \text{ rejects } v \mid T) = \Pr(\mathcal{A} \text{ rejects } v \mid T^{\text{sub}}).$$

Observe that in the Decision Phase of T^{sub} , when the algorithm reaches node v (which is a leaf in T^{sub}), it accepts v with probability $\tilde{p}_v$ and rejects it with probability

$$\Pr(\mathcal{A} \text{ rejects } v \mid T^{\text{sub}}) = 1 - \tilde{p}_v.$$

In contrast, for the original tree T , when the algorithm traverses the local subtree $T_v = \{v\} \cup \mathcal{C}(v)$, the probability of rejecting all nodes in T_v is:

$$\begin{aligned} & \mathbb{E}_{\mathcal{C}(v)} \Pr(\mathcal{A} \text{ rejects } v \mid T) \\ &= \mathbb{E}_{\mathcal{C}(v)} \left[\prod_{u \in \mathcal{C}(v)} (1 - \tilde{p}_u) \cdot (1 - \tilde{p}_v^{\text{res}}) \right] \\ &= \mathbb{E}_{\mathcal{C}(v)} \left[\frac{\left(1 - \sum_{u \in \mathcal{C}(v)} p_v(u)\right) \cdot p_v(\neg v)}{1 - \sum_{u \in \mathcal{C}(v)} p_v(u)} \right] \\ &= \mathbb{E}_{\mathcal{C}(v)} [p_v(\neg v)]. \end{aligned}$$

To avoid sacrificing the generality of this proof by using specific expressions (e.g., (4)–(7)), we solely rely on the identity $p_v(\neg v) = 1 - \sum_{u \in \Sigma} p_v(u)$ here. Together with the locally lossless property, we have

$$\mathbb{E}_{\mathcal{C}(v)} [p_v(\neg v)] = 1 - \sum_{u \in \Sigma} \mathbb{E}_{\mathcal{C}(v)} [p_v(u)] = 1 - \tilde{p}_v \sum_{u \in \Sigma} \mathcal{M}_b(u | v) = 1 - \tilde{p}_v.$$

Therefore, by the case analysis above, Equation (14) holds. Combining this with the observation that $\mathbb{E}_T[\cdot] = \mathbb{E}_{T^{\text{sub}}} [\mathbb{E}_{\mathcal{C}(v)}[\cdot]]$, we can establish:

$$\begin{aligned} & \mathbb{E}_T \left[\sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' | T]}{\mathcal{M}_b(o')} \right] \\ &= \mathbb{E}_{T^{\text{sub}}} \mathbb{E}_{\mathcal{C}(v)} \left[\sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' | T]}{\mathcal{M}_b(o')} \right] \\ &\stackrel{(14)}{=} \mathbb{E}_{T^{\text{sub}}} \left[\sum_{o' \subseteq o} \frac{\mathcal{M}_b(o) \Pr[\mathcal{A} = o' | T^{\text{sub}}]}{\mathcal{M}_b(o')} \right] \stackrel{(13)}{=} \mathcal{M}_b(o). \end{aligned}$$

□

B.3 Conditional Optimality of UniVer

Proof of Theorem 5. Given any possible Children nodes $\mathcal{C}(v) \sim \mathcal{M}_s^\neg(\cdot | v)$ of the non-leaf node v , the (conditional) acceptance rate for $\mathcal{C}(v) = \{u_1, \dots, u_m\}$ is

$$\begin{aligned} & \alpha_{\text{UniVer}}^*(\tilde{p}_v \mathcal{M}_b(\cdot | v), \mathcal{M}_s(\cdot | v)) \\ &= \mathbb{E}_{\mathcal{C}(v)} \left[\sum_{j=1}^m p_v(u_j) \right] = \sum_{j=1}^m \mathbb{E}_{\mathcal{C}(v)} [p_v(u_j)]. \end{aligned}$$

Note that for the first $\text{Top}_{m-1}(\mathcal{M}_s(\cdot | v))$ tokens $t \in H_v := \{u_1, \dots, u_{m-1}\}$, the expectation

$$\mathbb{E}_{\mathcal{C}(v)} [p_v(t)] = \tilde{p}_v \mathcal{M}_b(t | v)$$

always holds by the local losslessness of T . For the sample node u_m , via Eq. (4),

$$\begin{aligned} \mathbb{E}_{\mathcal{C}(v)} [p_v(u_m)] &= \sum_{u_m \in \Sigma} \mathcal{M}_s^\neg(u_m | v) p_v(u_m) \\ &= \sum_{u \in \Sigma} \min\{\tilde{p}_v \mathcal{M}_b(u | v), \mathcal{M}_s^\neg(u | v)\}. \end{aligned}$$

Therefore, the acceptance rate of UniVer equals to Eq. (10).

Obviously, under the constraint of local losslessness, the acceptance rate $\mathbb{E}_{\mathcal{C}(v)} [p_v(t)] = \tilde{p}_v \mathcal{M}_b(t | v)$ achieves the optimal upper bound for $t \in H_v$. In order to obtain the optimal acceptance rate $\mathbb{E}_{\mathcal{C}(v)} [p_v(u_m)]$ of the sample node u_m , we need to solve the following scaled optimal transport problem (Scaled-OT):

$$\begin{aligned} & \max \sum_{u \in \Sigma} \mathcal{M}_s^\neg(u | v) p_v(u) \\ & \text{s.t.} \quad \sum_{u \in \Sigma} \mathcal{M}_s^\neg(u | v) p_v(t) = \tilde{p}_v \mathcal{M}_b(t | v), \quad \forall t \in \Sigma. \end{aligned}$$

Equivalently, letting $\gamma_v(u, t) := \mathcal{M}_s^-(u \mid v)p_v(t)$, $\forall u \in \Sigma, t \in \Sigma^+ := \Sigma \cup \{\neg v\}$, the Scaled-OT problem is

$$\begin{aligned}
& \max \sum_{u \in \Sigma} \gamma_v(u, u) \\
& \text{s.t.} \quad \sum_{u \in \Sigma} \gamma_v(u, \neg v) = 1 - \tilde{p}_v, \\
& \quad \sum_{u \in \Sigma} \gamma_v(u, t) = \tilde{p}_v \mathcal{M}_b(t \mid v), \quad \forall t \in \Sigma, \\
& \quad \sum_{t \in \Sigma^+} \gamma_v(u, t) = \mathcal{M}_s^-(u \mid v), \quad \forall u \in \Sigma, \\
& \quad \gamma_v(u, t) \geq 0, \quad \forall (u, t) \in \Sigma \times \Sigma^+.
\end{aligned} \tag{15}$$

For the scaled optimal transport problem (15), the objective is to maximize the total mass on the diagonal $\sum_{u \in \Sigma} \gamma_v(u, u)$. By the marginal constraints, for each $u \in \Sigma$ we have:

$$\gamma_v(u, u) \leq \sum_{t \in \Sigma^+} \gamma_v(u, t) = \mathcal{M}_s^-(u \mid v), \tag{16}$$

and similarly for the column marginal:

$$\gamma_v(u, u) \leq \sum_{u' \in \Sigma} \gamma_v(u', u) = \tilde{p}_v \mathcal{M}_b(u \mid v). \tag{17}$$

Therefore,

$$\gamma_v(u, u) \leq \min\{\mathcal{M}_s^-(u \mid v), \tilde{p}_v \mathcal{M}_b(u \mid v)\}$$

for each u , yielding the upper bound:

$$\sum_{u \in \Sigma} \gamma_v(u, u) \leq \sum_{u \in \Sigma} \min\{\mathcal{M}_s^-(u \mid v), \tilde{p}_v \mathcal{M}_b(u \mid v)\}. \tag{18}$$

This bound is achieved by our UniVer and then we complete the proof of this conditional optimality theorem. $\square$

B.4 Superiority of UniVer over Greedy method

We first establish a more general lemma for the **modified UniVer** where the root node r is assigned an arbitrary effective acceptance probability $\tilde{p}_r \in [0, 1]$. The standard UniVer corresponds to $\tilde{p}_r = 1$.

Lemma 1 (Scaled Superiority). *For any draft tree T with root r and effective probability $\tilde{p}_r \in [0, 1]$, let $N_{\text{Mod}}(T)$ and $N_{\text{Greedy}}(T)$ denote the acceptance lengths of **modified UniVer** and Greedy method, respectively. Then the following superiority holds:*

$$\mathbb{E}_T[N_{\text{Mod}}(T)] \geq \tilde{p}_r \cdot \mathbb{E}_T[N_{\text{Greedy}}(T)]. \tag{19}$$

Proof of Lemma 1. We proceed by induction on the number of parent nodes n in T .

Base case ($n = 1$): The tree consists of only the root r and its children $\mathcal{C}(r)$. By Theorem 5 (Conditional Optimality):

$$\begin{aligned}
\mathbb{E}_T[N_{\text{Mod}}(T)] &= \alpha_{\text{UniVer}}^*(\tilde{p}_r \mathcal{M}_b(\cdot \mid r), \mathcal{M}_s(\cdot \mid r)) \\
&\geq \tilde{p}_r \cdot \alpha_{\text{Greedy}}^*(\mathcal{M}_b(\cdot \mid r), \mathcal{M}_s(\cdot \mid r)) \\
&= \tilde{p}_r \cdot \mathbb{E}[N_{\text{Greedy}}(T)],
\end{aligned}$$

where the inequality follows from the formulas of the acceptance rate (Eq. (2) and Eq. (10) for $\tilde{p} \in [0, 1]$).

Inductive step: Assume the lemma holds for all trees with at most k parent nodes. Consider a tree T with $k + 1$ parent nodes ($k \geq 1$). Let $\mathcal{C}(r) = \{v_1, \dots, v_m\}$ be the children of the root, with v_m being the sampled node. Since $n = k + 1 \geq 2$, at least one v_j is a non-leaf parent node. Without loss of generality, we let v_j be the first non-leaf node in the $\mathcal{C}(r)$ (i.e., v_j has descendants, while $v_1, \dots, v_{j-1}$ are leaves).

Define:

- T_{v_j} : the subtree rooted at v_j (including v_j and all its descendants);
- T^{new} : the tree obtained from T by removing all descendants of v_j (making v_j a leaf node).

Note that T_{v_j} and T^{new} both have at most k parent nodes, satisfying the induction hypothesis.

By the law of total expectation, conditioning on whether nodes $v_1, \dots, v_j$ are accepted or rejected, we decompose the expected acceptance length of modified UniVer as:

$$\begin{aligned}
\mathbb{E}_T[N_{\text{Mod}}(T)] &= \mathbb{E}_{T^{\text{new}}} \mathbb{E}_{T_{v_j} \setminus \{v_j\}}[N_{\text{Mod}}(T)] \\
&= \sum_{i=1}^{j-1} \mathbb{E}_{T^{\text{new}}} [\Pr(\text{accept } v_i) \cdot 1] \\
&\quad + \mathbb{E}_{T^{\text{new}}} \left[\Pr(\text{reach } v_j) \left(\tilde{p}_{v_j} + \mathbb{E}_{T_{v_j}}[N_{\text{Mod}}(T_{v_j}) \mid v_j] \right) \right] \\
&\quad + \mathbb{E}_{T^{\text{new}}} [\Pr(\text{reject } \{v_i\}_{i=1}^j) \cdot N_{\text{Mod}}(T^{\text{new}} \setminus \{v_i\}_{i=1}^j)] \\
&= \mathbb{E}_{T^{\text{new}}} [N_{\text{Mod}}(T^{\text{new}})] + \mathbb{E}_{T^{\text{new}}} \left[\Pr(\text{reach } v_j) \cdot \mathbb{E}_{T_{v_j}}[N_{\text{Mod}}(T_{v_j}) \mid v_j] \right].
\end{aligned} \tag{20}$$

where:

- $\Pr(\text{accept } v_i) = \prod_{\ell=1}^{i-1} (1 - \tilde{p}_{v_\ell}) \cdot \tilde{p}_{v_i} = p_r(v_i)$ is the marginal probability of accepting v_i ;
- $\Pr(\text{reach } v_j) = \prod_{\ell=1}^{j-1} (1 - \tilde{p}_{v_\ell})$;
- $\Pr(\text{reject } v_1, \dots, v_j) = \prod_{\ell=1}^j (1 - \tilde{p}_{v_\ell})$;
- $\mathbb{E}_{T_{v_j}}[N_{\text{Mod}}(T_{v_j}) \mid v_j]$ is the expectation acceptance length of modified UniVer with a draft tree T_{v_j} rooted at v_j .

Note that T^{new} contains at most k parent nodes (since v_j is now a leaf), and T_{v_j} also contains at most k parent nodes (since the original root $r \notin T_{v_j}$). Applying the induction hypothesis to both subtrees:

$$\mathbb{E}_{T_{v_j}}[N_{\text{Mod}}(T_{v_j}) \mid v_j] \geq \tilde{p}_{v_j} \cdot \mathbb{E}_{T_{v_j}}[N_{\text{Greedy}}(T_{v_j}) \mid v_j], \tag{21}$$

$$\mathbb{E}_{T^{\text{new}}}[N_{\text{Mod}}(T^{\text{new}})] \geq \tilde{p}_r \cdot \mathbb{E}_{T^{\text{new}}}[N_{\text{Greedy}}(T^{\text{new}})]. \tag{22}$$

Substituting (21) and (22) into (20), we obtain:

$$\mathbb{E}_T[N_{\text{Mod}}(T)] \geq \tilde{p}_r \cdot \mathbb{E}_{T^{\text{new}}}[N_{\text{Greedy}}(T^{\text{new}})] + \mathbb{E}_{T^{\text{new}}} \left[p_r(v_j) \cdot \mathbb{E}_{T_{v_j}}[N_{\text{Greedy}}(T_{v_j}) \mid v_j] \right]. \tag{23}$$

To complete the proof, we analyze the two cases based on whether v_j is a deterministic Top_{m-1} node or the sampled node v_m .

Case 1: $v_j \in \{v_1, \dots, v_{m-1}\}$ is a **deterministic node**. Since v_j belongs to the deterministic top- $(m-1)$ set, its value is fixed and independent of the randomness in T^{new} . Consequently, the subtree T_{v_j} is deterministic conditioned on v_j , making $\mathbb{E}_{T_{v_j}}[N_{\text{Greedy}}(T_{v_j}) \mid v_j]$ a constant that can be factored out of the expectation over T^{new} . By local losslessness, $\mathbb{E}_{T^{\text{new}}}[p_r(v_j)] = \tilde{p}_r \mathcal{M}_b(v_j \mid r)$, which equals $\tilde{p}_r \mathbb{E}_{T^{\text{new}}}[p_r^{\text{Greedy}}(v_j)]$ since Greedy method also accepts v_j with expectation $\mathbb{E}_{T^{\text{new}}}[p_r^{\text{Greedy}}(v_j)] = \mathcal{M}_b(v_j \mid r)$. Thus we can factor out/in the constant $\mathbb{E}_{T_{v_j}}[N_{\text{Greedy}}(T_{v_j}) \mid v_j]$ from the second term of (23), yielding:

$$\begin{aligned}
& \mathbb{E}_T[N_{\text{Mod}}(T)] \\
& \geq \tilde{p}_r \mathbb{E}_{T^{\text{new}}}[N_{\text{Greedy}}(T^{\text{new}})] + \tilde{p}_r \mathcal{M}_b(v_j \mid r) \cdot \mathbb{E}_{T_{v_j}}[N_{\text{Greedy}}(T_{v_j}) \mid v_j] \\
& = \tilde{p}_r \mathbb{E}_{T^{\text{new}}}[N_{\text{Greedy}}(T^{\text{new}})] + \tilde{p}_r \mathbb{E}_{T^{\text{new}}} \left[p_r^{\text{Greedy}}(v_j) \mathbb{E}_{T_{v_j}}[N_{\text{Greedy}}(T_{v_j}) \mid v_j] \right] \\
& = \tilde{p}_r \cdot \mathbb{E}_T[N_{\text{Greedy}}(T)].
\end{aligned}$$

Case 2: $v_j = v_m$ is the sampled node. For any token $u \sim \mathcal{M}_s^-(\cdot \mid r)$, the acceptance probability in UniVer is $p_r(u) = \min\{1, \frac{\tilde{p}_r \mathcal{M}_b(u \mid r)}{\mathcal{M}_s^-(u \mid r)}\}$, while in Greedy it is $p_r^{\text{Greedy}}(u) = \min\{1, \frac{\mathcal{M}_b(u \mid r)}{\mathcal{M}_s^-(u \mid r)}\}$. Since $\tilde{p}_r \leq 1$, we have $p_r(u) \geq \tilde{p}_r \cdot p_r^{\text{Greedy}}(u)$ for all u . Therefore, the second term of (23) implies

$$\mathbb{E}_{T^{\text{new}}} [p_r(v_m) \cdot \mathbb{E}_{T_{v_m}}[N_{\text{Greedy}}(T_{v_m}) \mid v_j]] \geq \tilde{p}_r \mathbb{E}_{T^{\text{new}}} [p_r^{\text{Greedy}}(v_m) \cdot \mathbb{E}_{T_{v_m}}[N_{\text{Greedy}}(T_{v_m}) \mid v_j]].$$

Substituting into (23) gives $\mathbb{E}_T[N_{\text{Mod}}(T)] \geq \tilde{p}_r \cdot \mathbb{E}_T[N_{\text{Greedy}}(T)]$, completing the induction.

By induction, Lemma 1 holds for all $n \geq 1$. $\square$

Proof of Theorem 6. Setting $\tilde{p}_r = 1$ in Lemma 1 yields exactly $\mathbb{E}[N_{\text{UniVer}}(T)] \geq \mathbb{E}[N_{\text{Greedy}}(T)]$, establishing the superiority of UniVer over Greedy method. $\square$